

Experiment/Practical 10 Ensemble Methods

Title: Implementation of Ensemble Methods for Classification

Aim: To implement and understand the concept of ensemble methods for classification tasks and evaluate their performance on a given dataset.

Objective: Students will learn

- The basic concept of ensemble learning and how it improves classification performance.
 - Various ensemble methods, including bagging, boosting, and stacking.
 - Implementing popular ensemble algorithms such as Random Forest, AdaBoost, and Gradient Boosting.
 - Evaluating the model's performance using relevant metrics such as accuracy, precision, recall, and F1-score.
 - Comparing the performance of ensemble models with individual classifiers.
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Problem statement

Use the given dataset(s) to demonstrate the application of ensemble methods for classification tasks. The task is to improve the classification performance by combining multiple weak classifiers to create a strong classifier using ensemble techniques.

Explanation/Stepwise Procedure/ Algorithm:

- Give a brief description of the ensemble techniques /methods.

Ensemble learning is a machine learning approach where multiple models called base learners or weak learners are combined to produce a more accurate and robust predictive model. Instead of relying on a single model, ensemble methods aggregate predictions from multiple models to reduce variance, bias or improve generalization. The fundamental idea is that a group of weak learners can come together to form a strong learner. In this experiment, ensemble techniques such as Random Forest (Bagging), AdaBoost (Boosting) and Gradient Boosting are implemented to improve classification performance.

- Describe types of ensemble techniques (Bagging, Boosting, Stacking).

a) Bagging (Bootstrap Aggregating):

- Multiple subsets of the dataset are created using random sampling with replacement.
- Each subset is used to train an independent model.
- Final prediction is made by averaging (for regression) or majority voting (for classification).
- Example: Random Forest

b) Boosting :

- Models are trained sequentially.
- Each new model focuses on correcting errors made by previous models.
- Assigns higher weights to misclassified samples.

Example: AdaBoost / Gradient Boosting

c) Stacking:

- Combines multiple different models.
 - Outputs of base models are used as input to a meta-model.
 - Meta-model makes the final prediction.
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- Give mathematical formulation (Bagging, Boosting, Stacking).

Bagging-

$$H(x) = \frac{1}{n} \sum_{i=1}^n h_i(x)$$

Boosting(AdaBoost)-

$$H(x) = \sum_{t=1}^T \alpha_t h_t(x)$$

where:

- $h_t(x)$ = weak learner
- α_t = weight of learner

Stacking-

$$H(x) = g(h_1(x), h_2(x), \dots, h_n(x))$$

here:

- g = meta-model
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- Write the importance of the ensemble techniques in the classification task
 - Improves model accuracy.
 - Reduces overfitting (especially bagging).
 - Handles complex data patterns.
 - Increases robustness.
 - Combines strengths of multiple models.
 - Mention applications of ensemble techniques
 - Fraud detection
 - Medical diagnosis
 - Recommendation systems
 - Image classification
 - Spam detection
 - Brief explanation of performance metrics (e.g., accuracy, precision, recall, f1-score, confusion matrix, cross-validation).

Accuracy:

$$\text{Accuracy} = \frac{TP + TN}{FP + FN + TP + TN}$$

Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Recall:

$$\text{Recall} = \frac{TP}{FN + TP}$$

F1 Score:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

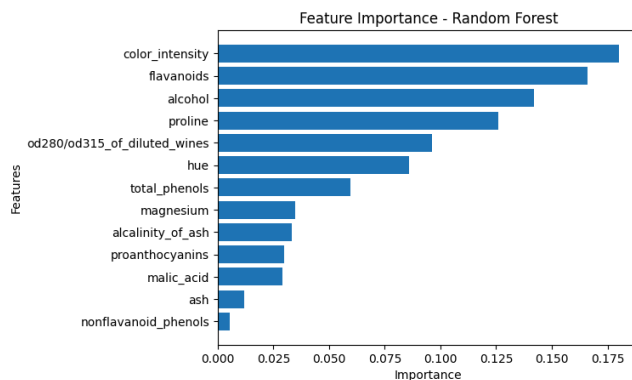
Confusion Matrix:

- Table showing TP, TN, FP, FN
- Helps visualize classification performance

Cross Validation:

- Data is split into multiple folds
- Model is trained and validated multiple times
- Provides reliable performance estimate

- **Add necessary figure(s)/Diagram(s)**

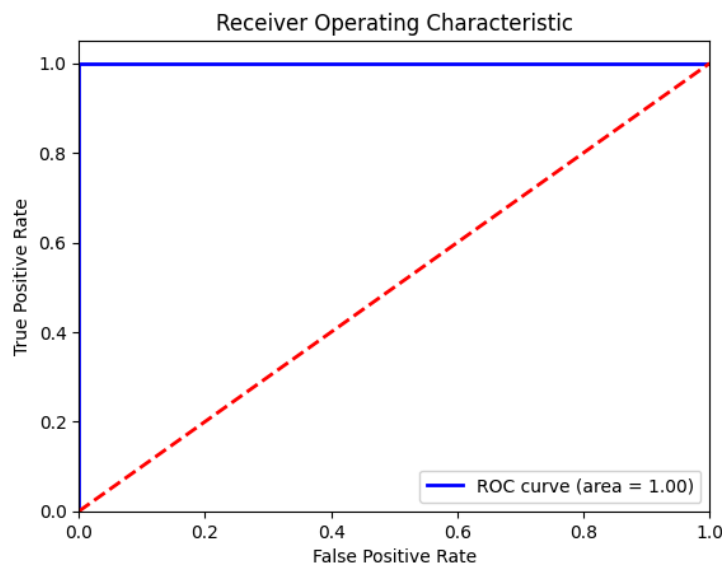


Input & Output:

Input:

- Dataset: Wine Dataset and Fashion_mnist
- User Input: Training and Testing data.
- Output:

- Predictions: The trained ensemble models generate predicted class labels for the test dataset. These predictions indicate the category or class assigned to each test instance based on learned patterns from training data.
- Model Evaluation Metrics: Accuracy, precision, recall, F1-score, confusion matrix, etc.
- Visualizations: Performance evaluation graphs (e.g., ROC curve, confusion matrix heatmap, performance comparison between models).



Conclusion:

Impact of Hyperparameters-

a) Number of Base Models ($n_{\text{estimators}}$):

- More models $\rightarrow$ better accuracy (up to a limit)
- Too many $\rightarrow$ increased computation

b) Learning Rate (Boosting):

- Controls contribution of each model
- Low value $\rightarrow$ better generalization but slower learning

c) Tree Depth:

- Controls model complexity
- High depth $\rightarrow$ overfitting
- Low depth $\rightarrow$ underfitting

Comparison of Ensemble Models:

Model	Technique	Strength	Weakness
Random Forest	Bagging	Reduce Variance	Can be slow
Ada Boost	Boosting	Focuses on difficult samples	Sensitive to noise
Gradient Boost	Bagging	High Accuracy	Computationally EXpensive